



Eco-friendly quaternary ammonium modified PDMS coatings with integrated anti-bacterial, anti-algae, and marine anti-fouling properties

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ABSTRACT

As the ecological risks associated with the use of copper and booster biocides in anti-fouling coatings become increasingly apparent, the design and development of environmentally friendly marine anti-fouling coatings present novel challenges. In this study, we developed a polyurethane (PU) modified polydimethylsiloxane (PDMS) copolymer (PDMS-PU) and then quaternized the copolymer using four brominated alkanes with varying carbon chain lengths. The soft segment PDMS, hard segments *N*-methylololamine (MDEA) and isophorone diisocyanate (IPDI), and strong cross-linked hydrogen bonds ensure the PDMS-PU polymer coatings with enhanced mechanical properties, good durability and static anti-algae properties. The quaternization of the coating resulted in a significant enhancement in its anti-bacterial and anti-algae performance. The modified coatings exhibited 99.1 % anti-bacterial inhibition against *Staphylococcus aureus* and *Escherichia coli*, and achieved 88.9 % and 80.2 % anti-algae inhibition rate for *Chlorella* and *P. tricornutum*, respectively. The anti-algae mechanism involves an increase in membrane potential followed by rupture, resulting in cellular damage and DNA leakage. In addition, the PDMS-PU-X coatings (where X = 4, 8, 12, 16, representing the number of carbon atoms in the bromoalkanes used for quaternization) demonstrated excellent anti-fouling performance against marine microorganisms, liquid pollutants, and pseudo-barnacle adhesion. The laboratory zebrafish toxicity tests further confirmed the coatings' non-toxicity toward non-target organisms. This study provides a novel strategy for developing environmentally friendly marine anti-fouling coatings.

1. Introduction

The issue of marine bio-fouling, which has a significant impact on functionality of the marine equipment surfaces, has become increasingly prevalent in line with the expansion of the marine industry [1–3]. Previously, biocides were frequently employed in the formulation of anti-fouling paints. However, these conventional agents were found to be detrimental to marine life and the wider ecosystem. Due to its toxicity, tributyltin (TBT), a widely used traditional biocide, was prohibited by the International Maritime Organization in 2003 [4]. Consequently, copper-based coatings were developed as an alternative solution, designed to slowly release copper in order to prevent fouling. However,

concerns about the environmental impact of leached copper have led to increasing restrictions and bans on these coatings [5]. This has driven modern research efforts to develop environmentally friendly and durable marine anti-fouling coatings [6–8].

In recent years, polydimethylsiloxane (PDMS) coatings have emerged as a promising solution for environmentally friendly anti-fouling due to their low surface free energy and non-toxic properties [9,10]. In general, PDMS is a type of silicone polymer composed of a -Si-O- backbone with two methyl substitutes attached to each silicon atom. The flexibility of the siloxane backbone allows the chains to easily arrange and rearrange, positioning the methyl groups at the surfaces and interfaces, which results in low surface free energy [11,12]. The low

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surface free energy of PDMS reduces the adhesion strength of fouling organisms to the coating. Consequently, these organisms can be easily removed from the surface by external shear forces [13]. However, these coatings often exhibit from limited durability, mechanical properties, and poor adhesion to substrates [14,15].

To address the limitations of PDMS coatings, researchers have developed anti-fouling coatings based on siloxane-polyurethane (SiPU) systems [16]. These coatings demonstrate significant improvements in mechanical properties, adhesion, and durability, while also retaining effective foul-release (FR) characteristics [17]. The polyurethane (PU) segments form a hydrogen-bonded cross-linking network in the coating due to hydrogen bonding, which enhance the mechanical properties of the coating further [15,18,19]. By optimizing the ratio of its soft and hard segments, coatings with enhanced functionality can be produced [20,21]. For example, Chao Liu and colleagues developed a polyurethane-modified coating that exhibited enhanced adhesion strength (2.55 MPa) to the substrate due to hydrogen bonding of the urea groups [22]. However, in coastal areas and near harbors, the enrichment of maritime bacteria (both *Gram-negative* and *Gram-positive*) and algae (e.g., *diatoms*) often leads to fouling of PDMS coatings under static seawater immersion conditions (e.g., ships docked in harbors) [23]. Franck Cassé et al. exposed PDMS coatings to a 60-day static immersion period and observed that the surface became enriched with a substantial number of bacteria and diatoms, resulting in over 95.0 % of the coated surface being covered with fouling [24]. These findings indicate that the static anti-fouling performance of PDMS coatings is poor, prompting researchers to further enhance static fouling resistance [25–27].

Quaternary ammonium salts (QAS) have demonstrated strong anti-bacterial and anti-algae properties, which can be attributed to their ability to interact with negatively charged cell membranes through their positive charge [28–30]. Furthermore, cross-linked polysiloxane matrices with quaternary ammonium groups are also considered to be optimal materials for the development of anti-fouling and decontamination coatings [31,32]. Partha Majumdar et al. developed polydimethylsiloxanes with quaternary ammonium groups, achieving an 80.0 % reduction in biofilm retention and enhanced static anti-fouling properties of the coatings [33]. Similarly, Yingqing Chen et al. prepared an anti-algae coating for wastewater facilities by covalent compounding of quaternary ammonium salts with aqueous enamel coatings, achieving an inhibition rate of 78.2 % against *Chlorella* [34]. Despite the existence of studies that have sought to introduce quaternary ammonium salts into PDMS coatings, the mechanism of their inhibition of algae remains unclear.

In general, the anti-bacterial mechanism of quaternary ammonium salts involves the irreversible damage of cell membranes. This occurs through strong electrostatic interactions between the positively charged nitrogen atoms and the negatively charged cell membranes, along with interactions between the long-chain alkyl groups and the phospholipid bilayer of the cells [35]. However, algae present a greater challenge due to their more complex cell membrane and wall compositions, resulting in a less comprehensive understanding of the anti-algae mechanism of quaternary ammonium salts. To address this gap in knowledge, this study systematically investigates the anti-algae performance and underlying mechanisms of quaternary ammonium salts by synthesizing and testing four variants with distinct alkyl chain lengths. In this paper, the quantitative relationship between the alkyl chain length of these compounds and their anti-algae efficacy was elucidated through comparative analysis of the coatings' performance and mechanistic behavior. This provides a theoretical basis for the molecular design of anti-algae coatings.

In this study, we developed a polyurethane-modified PDMS and then quaternized the polymer using four brominated alkanes with varying carbon chain lengths. The urea and carbamate groups present in the hard segments of the polymer are capable of forming hydrogen bonds with the substrate, thereby significantly enhancing the adhesion

properties. Furthermore, the binding of these hard segments facilitates the ordered concentration of quaternary ammonium groups and their anchoring at the surface, thereby enhancing the anti-bacterial and anti-algae properties of the coating. Especially in response to the poor static anti-algae ability of PDMS coatings, the coating prepared in this paper has significant anti-algae adhesion and growth ability under static and dynamic conditions, which has not been mentioned in other literature. In contrast to commercial cuprous oxide (Cu_2O) coatings, this modified coating exhibited non-toxicity to zebrafish embryos, larvae, and adults, predominantly due to its mechanism of contact killing as opposed to the release of biocidal substances. The development of a PDMS-PU-X coating has been undertaken, which exhibits high adhesion, static/dynamic anti-algae performance and eco-friendly. This coating has been formulated to address the issue of the poor static anti-algae ability of traditional PDMS coating. Furthermore, it was elucidated that the anti-algae mechanism of quaternary ammonium salts is by affecting the membrane potential of algae cells, leading to an increase in membrane permeability. This, in turn, precipitates algae cell rupture, DNA leakage, and oxidative stress. These findings present a novel approach to developing eco-friendly and effective marine anti-fouling coatings, offering a theoretical basis for the structural design and performance optimization of quaternary ammonium salt-modified coatings.

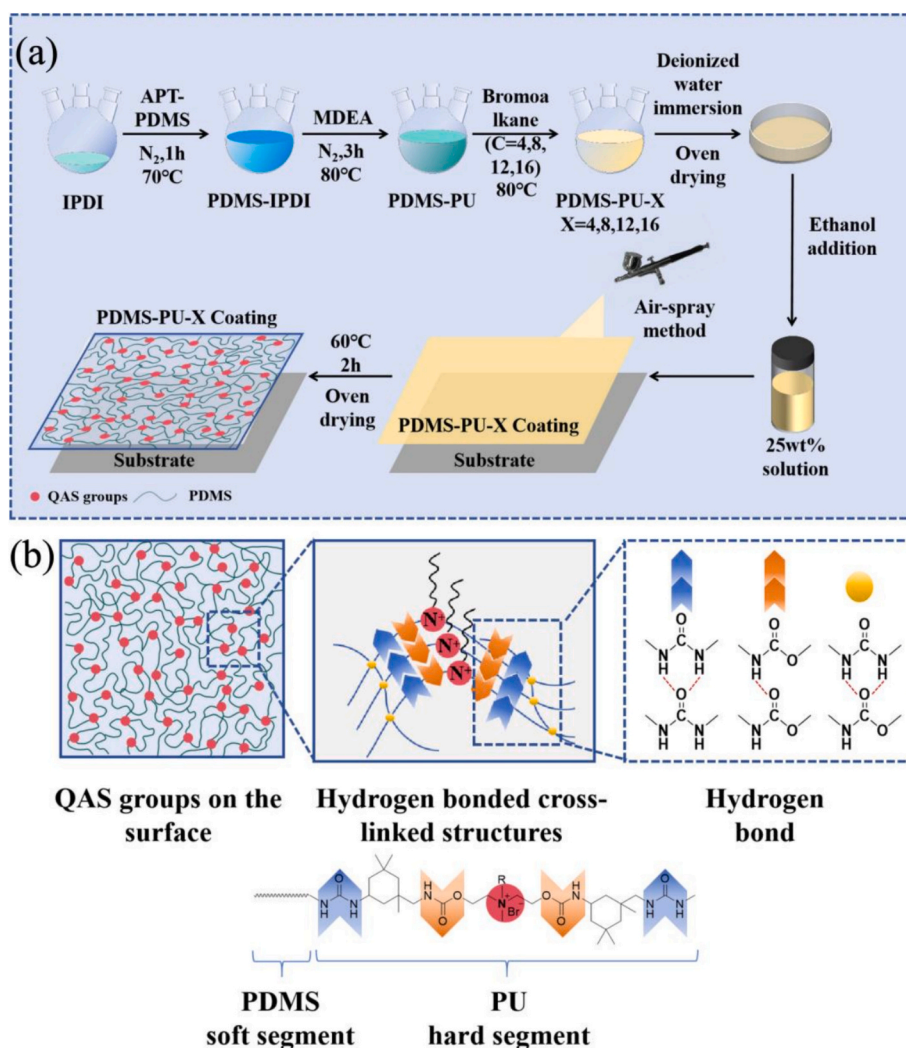
2. Experimental section

2.1. Materials and reagents

α , ω -aminopropyl terminated polydimethylsiloxane (APT-PDMS, molecular weight 1000 g/mol, 98.0 %) was procured from Hangzhou Silong Mater. Tech., China. Isophorone diisocyanate (IPDI, ≥ 99.0 %) was procured from Sigma-Aldrich. *N*-methyl-diethanolamine (MDEA, ≥ 99.0 %) was obtained from Xiya Reagent. 1-Bromobutane (≥ 99.5 %), 1-Bromooctane (≥ 98.0 %), 1-Bromododecane (≥ 98.0 %), as well as 1-Bromohexadecane (≥ 97.0 %), and 2', 7'-Dichlorodihydrofluorescein diacetate (DCFH-DA, ≥ 97.0 %) were obtained from Aladdin. The nucleic acid stain SYTOX green (99.0 %) was obtained from EnoGene. Acid Orange 7 (AO7, 98.4 %) was procured from Selleck. The Propidium Iodine (PI, ≥ 95.0 %) and Malondialdehyde (MDA) Content Assay Kit was sourced from Solarbio. The strains of algae, *Chlorella vulgaris* (XR003090) and *Phaeodactylum tricornutum* Bohlin (XR03195), were obtained from Fakanuo. The composition of the Algae Nutrient Solution is detailed in Table S4. DiBAC4 (3) (98.8 %) was procured from MCE, while PDMS-184 (≥ 99.9 %) was obtained from Dow Corning. The Cu_2O coating was sourced from Kailin New Materials Co., Ltd., China.

2.2. Preparation of PDMS-PU-X (X = 4, 8, 12, 16) coatings

The synthesis of the PDMS-PU-X (X = 4, 8, 12, 16) coatings was conducted according to the procedure illustrated in Scheme 1a. The initial step involved capping the APT-PDMS with an excess of IPDI, thereby forming a prepolymer. Subsequently, a stoichiometric quantity of chain extender was added in a dropwise manner. Specifically, 3.33 g (15.0 mmol) of IPDI was dissolved in 5.0 mL of dried toluene and added to a three-necked round-bottom flask equipped with a condenser, dropping funnel, nitrogen gas inlet, and magnetic stirrer bar. APT-PDMS (7.5 mmol) was dissolved in 60.0 mL of dried toluene, and this solution was added dropwise under a nitrogen atmosphere over the course of one hour. Subsequently, the mixture was stirred for an additional hour at 70 °C. Subsequently, 0.83 g (7.0 mmol) of *N*-methyl-diethanolamine (MDEA) in 5.0 mL of dried toluene was added dropwise to the flask. The mixture was then stirred for a further 3 h under nitrogen at 80 °C, resulting in the product designated as PDMS-PU. The quaternization of PDMS-PU was conducted at 80 °C by the addition of 1-bromobutane, 1-bromooctane, 1-bromododecane or 1-bromohexadecane, respectively. Following an additional 8 h of reaction, the reactants were dried in an oven and soaked subsequently in deionized water for 24 h in order to



Scheme 1. (a) Preparation of PDMS-PU-X coatings and (b) Schematic representation of soft and hard segments and hydrogen-bonded cross-linked structures.

remove any unreacted small molecules. For simplicity, the resulting products were labelled as PDMS-PU-X ($X = 4, 8, 12, 16$), where X represents the number of carbon atoms in the bromoalkanes. The reaction equations are shown in Fig. S3.

The aluminum sheet substrate was sequentially abraded using sandpaper to eliminate the native oxide layer. Subsequently, the substrate was subjected to plasma treatment for 5 min using a plasma cleaner (YX-CR800-R, Dongguan YiXing Intelligent Technology Co., China). Specific treatment procedures and devices can be found in Supplementary Material.

Subsequently, a solution of PDMS-PU-X in ethanol was prepared, with a concentration of 25.0 wt%. This solution was uniformly sprayed onto the aluminum sheet surface using a spray gun set at an air pressure of 1.5 bar. Finally, the coating was dried and cured in an oven at 60 °C for 2 h.

2.3. Characterization of PDMS-PU-X coatings

The chemical compositions of the coatings were determined by nuclear magnetic resonance spectroscopy (NMR, Avance NEO 400 MHz, Bruker, Germany) and FTIR spectroscopy (Nicolet 6700, Thermo, USA). The molecular weight of the polymer is measured by the Gel Permeation Chromatography (GPC, Agilent 50, USA).

2.4. Degrees of the charge density of the PDMS-PU-X coatings

The degrees of charge density (CD) of the PDMS-PU-X were determined by ultraviolet/visible spectroscopy via a colorimetric method based on fluorescent complexation, as previously described [36]. It was assumed that each quaternary ammonium group of the polymer interacted with a single dye molecule in order to convert the dye concentration, determined by UV/vis spectroscopy, to the charge density of the polymer brush. Specific experimental steps are described in the Supplementary Material. The degrees of charge density of polyurethane were calculated using the following equation (Eq. (1)).

$$CD = (VCN_A)/A \quad (1)$$

where V is the volume of the extraction solution, N_A is the Avogadro number, and A is the surface area of the polymer samples.

2.5. Anti-bacterial properties of the PDMS-PU-X coatings

Bacteria have been found to be enriched in coastal and harbor areas due to the increased movement of people in these areas, which has been shown to accelerate the fouling of PDMS coatings [23]. Of particular concern are *Gram-negative Escherichia coli* (*E. coli*) and *Gram-positive Staphylococcus aureus* (*S. aureus*), which are readily identified in such environments [37,38]. Consequently, these two bacterial species have been selected for further study as bacterial model [39–41]. Specific steps

of the anti-bacterial experiments are described in the Supplementary Material. Each coating composition was subjected to three anti-bacterial tests.

The potential for leaching of active quaternary ammonium compound (QAC) species from the films was investigated through the utilization of a conventional zone-of-inhibition test. Each coating composition was subjected to three anti-bacterial tests.

2.6. Anti-algae properties of the PDMS-PU-X coatings

The anti-algae performance of the PDMS-PU-X coatings was evaluated using *Chlorella* and *Phaeodactylum tricornutum* as model organisms, referencing the standard (China, GB/T 21353-2008). In summary, aluminum sheets ($3 \times 3 \text{ cm}^2$) coated with PDMS-PU-X were placed in separate beakers, rinsed with deionized (DI) water to remove unbound residues, and sterilized under UV light for 30 mins. A suspension of algae cultured in algae nutrient solution (10^7 cells/mL) was added to each beaker at a volume of 20.0 mL. The uncoated aluminum sheet was designated as the control (as shown in Table S5). Subsequently, the beakers were covered with cling film and placed in the thermostatic incubator (MQT-50, Minquan, China), where they were illuminated with a lamp to adjust the light intensity to a range of 1000-1200 LUX. They were then incubated at room temperature (25°C) for 14 days on a 12:12 light-dark cycle. The samples were initially rinsed with deionized (DI) water at 20 kPa to remove loosely adhered algae. Subsequently, the samples were placed in separate beakers and sonicated for 10 min to disrupt any strongly adhered algae (Sonographer, SK250LHC, KuDos, China). After sonication, the samples were rinsed again with DI water for 1 min using a wash bottle. Specific flushing procedures and devices can be found in Supplementary Material and Video S1. The algae collected in this manner were then used in a series of experiments. Concurrently, the investigation was undertaken to ascertain whether the coating retained its anti-algae properties when submerged in a seawater environment. The PDMS-PU-X coating was subjected to continuous immersion in a 3.5 wt% NaCl solution for a period of 30 days, after which a re-evaluation of the coating's anti-algae properties was conducted. Firstly, the optical density (OD) of the collected algae suspension was determined at 680 nm (*Chlorella*, OD_{680}) and 540 nm (*P. tricornutum*, OD_{540}), respectively, using a UV/vis spectrophotometer (Perkin-Elmer LAMBDA 950) to quantify the amount of attached algae. Each coating composition was subjected to three anti-algae tests. The anti-algae resistance rate of coating was calculated using the following equation (Eq. (2)).

$$\text{Anti-algae resistance rate(\%)} = ((\text{OD}_{\text{PU}} - \text{OD}_X) / \text{OD}_{\text{PU}}) \times 100\% \quad (2)$$

where OD_X is the optical density value of algae on the PDMS-PU-X, and OD_{PU} is the optical density value of algae on the PDMS-PU.

Secondly, confocal laser scanning microscopy (CLSM, LSM 900, Zeiss, Germany) was employed to differentiate between viable and non-viable cells of the algae. The SYTOX green nucleic stain was observed to permeate cells with poor membrane integrity, in accordance with the staining procedure described by Veldhuis et al [42]. The fluorescence of the SYTOX green nucleic stain (excitation 488 nm, emission 515 nm) and the auto-fluorescence of chlorophyll-a (excitation 640 nm, emission 590 nm) were recorded. To provide a quantitative assessment of the CLSM results, the viable algae ratio (calculated as the area of the green signal divided by the area of the red signal) was determined for the coated samples using the ImageJ software. The algae inhibition rate of coating was calculated using the following equation (Eq. (3)).

$$\text{Algae inhibition rate(\%)} = (A_G / A_R) \times 100\% \quad (3)$$

where A_G is the area of the green signal, and A_R is the area of the red signal.

Thirdly, the detection of reactive oxygen species (ROS) was conducted using a commercially available kit designed for this purpose. In brief, $5.0 \mu\text{M}$ of DCFH-DA was combined with the algae sample (1.0 mL)

and incubated in the dark for 20 mins at 37°C . The mixture was then washed with a PBS solution in order to remove any residual probes. The fluorescence intensity of DCF, which is the oxidation product of DCFH, was subsequently detected using fluorescence spectroscopy (Fluorolog-3, Horiba, USA) at an excitation wavelength of 488 nm and an emission wavelength of 525 nm.

Subsequently, the collected algae suspension was subjected to centrifugation, and the resulting supernatant was obtained. The DNA fraction in the supernatant was isolated using a DNA purification kit (Beyotime, China) and subsequently analyzed by fluorescence spectroscopy (Fluorolog-3, Horiba, USA) using propidium iodide (PI, $10.0 \mu\text{g/mL}$) as a probe (excitation/emission wavelengths 535/615 nm).

Finally, the extent of lipid peroxidation of the cell membranes was determined by quantifying the malondialdehyde (MDA) content in accordance with the instructions provided with a specific kit from Solarbio. The voltage-sensitive dye DiBAC4 (3) was employed to observe alterations in membrane potential. In brief, DiBAC4 (3) at a final concentration of $10.0 \mu\text{M}$ was combined with the algae sample (1.0 mL) and incubated in the dark for 30 mins at 37°C . The samples were imaged using confocal laser scanning microscopy (CLSM, LSM 900, Zeiss, Germany). The morphology of the attached algae cells on the coating surface was further visualised by SEM (MIRA4, Tescan).

2.7. Anti-fouling properties

The algae *Chlorella* and *Phaeodactylum tricornutum* were selected as the model organisms for the anti-fouling experiment [43]. The coatings were applied to glass slides using a paintbrush to ensure a uniform layer. A suspension of algae cultured in algae nutrient solution (10^5 cells/mL) was added to each Petri dish at a volume of 10.0 mL. The coatings were incubated in a thermostatic incubator set at 25°C with the aforementioned algae suspension on a 12:12 light-dark cycle (the light intensity to a range of 1000-1200 LUX). Following a 24-h incubation period, the samples were photographed using an inverted fluorescence microscope (Olympus IX71, Japan). Each sample was randomly selected and observed under a fluorescence microscope to determine the quantity of algae that had attached. Specific flushing procedures and devices can be found in Video S1. Post-washing, the samples were re-examined under the inverted fluorescence microscope to determine the remaining algae count. Each coating composition was subjected to three anti-fouling tests. The adhesion strength of the fouling organisms was evaluated based on the removal rate, calculated using Eq. (4).

$$\text{Removal rate(\%)} = ((N_0 - N_1) / N_0) \times 100\% \quad (4)$$

where N_0 is the average amount of adhesion before scour, and N_1 is the average amount of adhesion after washing.

To further simulate the practical application scenarios of the coating, we investigated the dynamic anti-algae and anti-fouling performance of the coatings. Specifically, aluminum sheets ($8 \times 1 \text{ cm}^2$) coated with PDMS-PU-X were bonded to the sides of a model boat. The system was then immersed in a square glass tank ($23 \times 16 \times 25 \text{ cm}^3$) containing 6.0 L of algae suspension cultured in algae nutrient solution (10^5 cells/mL). The experimental setup included a 20.0 W water pump generating a flow rate of 1.0 m/s at the pump outlet. After 7 days of continuous operation, the coating surfaces were examined using an industrial optical microscope (GP-300C, GaoPin, China).

To illustrate the coating's adaptability to a multitude of practical applications, the samples were submerged in four distinct liquids, including milk, mud water, coffee, and a water solution dyed with Food Blue (CAS: 2650-18-2), and subsequently removed to ascertain whether any residual liquid remained on the surface.

The standard push-off assay, as outlined in ASTM D-5618 (2011), was employed to quantify the adhesion strength of pseudo-barnacles. Firstly, an aluminum stud (diameter 10 mm, height 10 mm) was bonded to the coatings using epoxy adhesive (Araldite 2014). The

adhesive was then cured for 12 h at 120 °C in an oven. The maximal shear force was subsequently measured using a force gauge (YHT NK-500, Shenzhen Yuanhengtong Co., Ltd).

2.8. Durability of PDMS-PU-X coatings

In accordance with the ASTM D4541 standard, the pull-off adhesion of the coating to the substrate was determined through the utilization of a tensile tester (SH-M, China). To further evaluate the durability of the PDMS-PU-X coatings, the resulting coating was immersed in a 3.5 wt% NaCl solution for a period of 30 days. Furthermore, the impact of thermal ageing (for 30 days) and the cyclic sand abrasion test on the surface of the PDMS-PU-X was investigated [43]. The surface hydrophobicity of the coatings was assessed by measuring the contact angle (CA) of water (Dataphysics OCA100). Before measuring the water contact angle of the sample, the surface was rinsed with deionized water three times immediately after removal, and it was necessary to go through a drying process at 100 °C for 12 h and use nitrogen to clean the surface of the sample before carrying out the contact angle measurement. The test method used was the solid drop method in an environment of 25 °C and 50.0 % relative humidity. The liquids used were deionized water (3.0 μ L) and diiodomethane (2.0 μ L). The polymer samples were purged with nitrogen prior to testing. Five different areas were tested for each sample and the average was taken as the contact angle measurement. The surface free energy of the polymer coatings was determined from the water contact angle and the diiodomethane (Aladdin, ≥ 98.0 %) contact angle according to the Owens-Wendt-Rabel-Kaelble method. Specific steps for the calculation of the surface free energy are given in the supplementary material. The specific equation, Eq. (5), is given below:

$$\gamma_L(1 + \cos\theta) = 2\left(\sqrt{\gamma_s^d \gamma_L^d} + \sqrt{\gamma_s^p \gamma_L^p}\right) \quad (5)$$

where γ_L , γ_L^D and γ_L^P are the total, dispersive, and polar surface energies of the liquid, respectively, θ is the contact angle.

2.9. Toxicity assay

Referencing the ISO 15088 standard, the toxicity of PDMS-PU-X and Cu₂O coatings was evaluated using zebrafish as an experimental model. Wild-type zebrafish embryos, juveniles, and adults were obtained from National Zebrafish Resource Centre. Newly laid eggs were siphoned from the tank in the morning. Embryos were monitored until they reached the blastula stage and then used in experiments. Experiments were initiated with eggs containing blastulae.

The PDMS-PU-X and Cu₂O coatings were prepared in 6-well polystyrene plates. For the experiment, embryos from the same day were pooled and randomly assigned (10 embryos per well) to each well containing the test coatings with 5.0 mL of embryo culture medium (ECM). The control group consisted of ten embryos placed in 6-well plates with 5.0 mL of ECM [44]. Specific steps for embryo cultures and devices are described in the Supplementary Material. The hatching success rate of the zebrafish embryo was calculated using the following equation (Eq. (6)).

$$\text{Hatching success rate}(\%) = (N_E/10) \times 100\% \quad (6)$$

where N_E is the number of successfully hatched zebrafish embryos.

The early-stage fish demonstrate robust responses to contaminants, exhibiting sensitivities and physiological mechanisms that are analogous to those observed in adults. However, the larval stage is the most sensitive, as the chorion affords embryos extra protection [45]. Consequently, the impact of the coating was subjected to further examination using zebrafish larvae. Specific steps for larval cultures and devices are described in the Supplementary Material. The larval survival rate of zebrafish was calculated using the following equation (Eq. (7)).

$$\text{Larval Survival rate}(\%) = (N_L/10) \times 100\% \quad (7)$$

where N_L is the the number of successfully surviving zebrafish larvae.

In order to evaluate the acute toxicity of the coating, it is necessary to refer to the standard (China, GB/T 13267-91) and use adult zebrafish. Specific steps for adult cultures and devices are described in the Supplementary Material. The survival of the adult zebrafish was monitored visually, and the adult survival rate of zebrafish embryos was calculated using the following equation (Eq. (8)).

$$\text{Adult Survival rate}(\%) = (N_A/5) \times 100\% \quad (8)$$

where N_A is the the number of successfully surviving adult zebrafish.

3. Results and discussion

3.1. Characterization of PDMS-PU-X polymer

Fig. 1a illustrates the FTIR spectra of APT-PDMS, PDMS-IPDI, and PDMS-PU. The PDMS-PU exhibited distinctive peaks at 1636 cm^{-1} and 1701 cm^{-1} , which correspond to the C=O stretch vibration absorption of the urethane and carbamate groups, respectively. The isocyanate peak observed at 2265 cm^{-1} in PDMS-IPDI, disappeared completely in the case of PDMS-PU, which suggests that the reaction between the excessive isocyanate and MDEA has been successful. The peak observed at 1260 cm^{-1} is the bending vibration absorption peak of Si – CH₃ in the PDMS segment, two strong characteristic absorption peaks at 1000–1100 cm^{-1} belong to the bending vibration absorption peak of Si – O – Si group in PDMS, and 806 cm^{-1} is the rocking vibration absorption peak of Si – CH₃ group in the PDMS segment. The peak observed at 2960 cm^{-1} is the C – H stretching vibration peak in CH₃. The absorption around 3320 cm^{-1} is presumably attributed to the N – H stretching vibrations [46]. In addition, a peak at 1559 cm^{-1} , attributed to N – H bending vibrations [46]. The GPC data (Table S6) indicate that the molecular weight of the prepolymer, PDMS-IPDI, is relatively low in comparison to that of the starting material, which suggests that the IPDI effectively capped the APT-PDMS. Subsequently, the molecular weight of PDMS-PU increased to 37,621 g/mol, indicating successful polymerization and chain extension.

Fig. 1b illustrates the FTIR spectra of PDMS-PU and PDMS-PU-X. All samples displayed a distinctive peak at 1636 cm^{-1} and 1701 cm^{-1} , indicative of the C=O stretch vibration absorption of the urethane and carbamate groups, respectively. The characterization of PDMS-PU-X revealed the presence of a new signal at 1470 cm^{-1} , which can be attributed to the quaternary ammonium (N⁺ – CH₃) groups [46]. The ¹H NMR spectrum of PDMS-PU-X is presented in Fig. S4a, measured in deuterated chloroform at room temperature (25 °C). Following the quaternization process, a new triplet peak emerges at $\delta = 3.38$, which is attributed to the N⁺-CH₂ protons, and another peak at $\delta = 3.28$ corresponds to the N⁺-CH₃ protons. Furthermore, the intensity of these peaks diminishes as the carbon chain length increases, potentially due to variations in the degree of quaternization of PDMS-PU-X. The ¹³C NMR spectrum of PDMS-PU-X is presented in Fig. S4b, measured in deuterated chloroform at room temperature (25 °C). Two distinct carbon peaks are observed in the spectrum of the MDEA: one at $\delta = 55.89$, attributed to carbon atoms bonded to nitrogen atoms, and another at $\delta = 42.07$, attributed to carbon atoms bonded to oxygen atoms. Subsequent formation of the quaternary ammonium group results in the appearance of additional peaks in the TPA spectrum. A new peak at $\delta = 64.20$ is attributed to the carbon on the ethyl group connected to the quaternary nitrogen ion (N⁺). The peak at $\delta = 55.89$ corresponds to carbon atoms bonded to oxygen, while the peak at $\delta = 50.58$ is assigned to the carbon on the methyl group connected to the N⁺ ion. The consistent peak positions between $\delta = 40$ –70 in the PDMS-PU-X spectra provide further confirmation of the successful quaternization of the tertiary nitrogen atoms along the PDMS-PU-X chain.

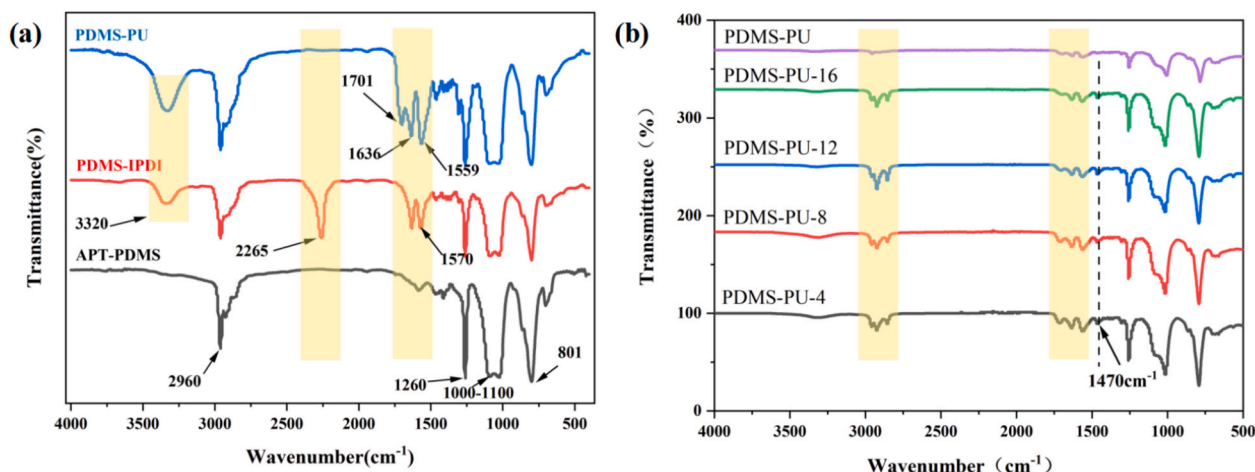


Fig. 1. (a) IR spectra of the APT-PDMS, PDMS-IPDI, and PDMS-PU (b) IR spectra of PDMS-PU and PDMS-PU-X.

The aforementioned results indicate that the successful quaternization of PDMS-PU-X has been achieved. To further investigate the surface properties of the coatings, XPS and charge density analysis of the PDMS-PU-X were performed.

3.2. The degrees of quaternization and the charge density of the PDMS-PU-X coatings

The degree of quaternization and the charge density (CD) of the coatings have been demonstrated to exert a significant influence on the anti-bacterial [47]. The surface chemical composition of the PDMS-PU-X coatings was analyzed by X-ray photoelectron spectroscopy (XPS). As shown in Fig. S5, PDMS-PU and PDMS-PU-X were predominantly composed of carbon (C), nitrogen (N), silicon (Si), and oxygen (O), with

characteristic binding energies observed at 284.8 eV (C 1s), 399.7 eV (N 1s), 99.8 eV (Si 2p), 152.8 eV (Si 2s), and 530.8 eV (O 1s), respectively. Notably, distinct Br3d signals were detected exclusively on the PDMS-PU-X surface, which can be attributed to the surface enrichment of quaternary ammonium salt. As illustrated in Fig. 2b-e, the high-resolution N 1s spectra of all coating samples can be resolved into two distinct peaks at binding energies of 399.3 and 401.0 eV, which are attributed to C—N and C—N⁺, respectively. A notable enhancement in the intensity of the C—N⁺ peak for PDMS-PU-X coatings in comparison to PDMS-PU (Fig. 2a) substantiates the successful quaternization of the coatings. As illustrated in Fig. 2b-e, the signal peak of N⁺—C diminished with the augmentation of alkyl chain lengths, thereby indicating a reduction in the degree of quaternization. The degree of quaternization (DQ) of coatings was determined using the following equation (Eq. (9)).

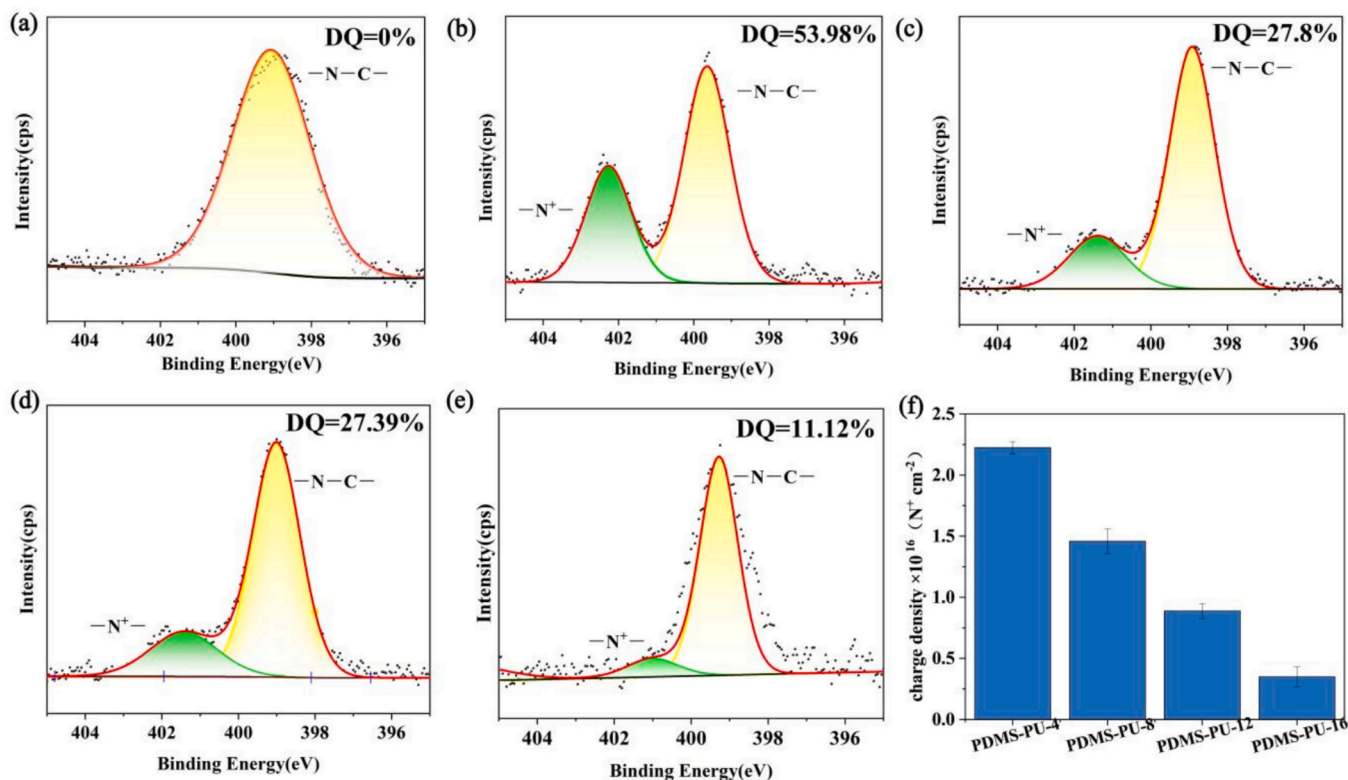


Fig. 2. High-resolution XPS N 1s spectra of the (a) PDMS-PU, (b) PDMS-PU-4, (c) PDMS-PU-8, (d) PDMS-PU-12, (e) PDMS-PU-16, and (f) The degrees of the charge density of the PDMS-PU-X coatings.

$$DQ(\%) = (A_1/A_N) \times 100\% \quad (9)$$

where A_N is the peak area for -N-C- peaks and A_1 is the peak area for -N⁺- peaks.

The reduction of the DQ with the augmentation of alkyl chain lengths maybe from the alkyl chain-induced steric hindrance which increased the difficulty of quaternization and reduced the effectiveness of quaternization. On the other hand, the curling of the alkyl chain can also affect the degree of quaternization.

The literature reports have established that the surface concentration of quaternary ammonium groups is a critical factor governing anti-bacterial efficacy, with charge density exceeding a threshold of $1.0 \times 10^{14} \text{ N}^+ \text{ cm}^{-2}$ required to achieve anti-bacterial activity [48]. As demonstrated in Fig. S6a, the UV/vis absorption spectra of the desorption solution exhibit a characteristic absorption band at 450-550 nm, with a maximum absorption at 483 nm. It is evident from the figure that the desorbed solution's absorbance diminishes with an increase in alkane chain length, signifying that the number of AO7 molecules dissociated from the coating surface is considerably reduced, and the cationic charge density on the coating surface decreases. In this study, the charge density of the PDMS-PU-X coatings were determined using a colorimetric method, with the specific data presented in Table S7. As shown in Fig. 2f, the charge density decreases with the growth of the alkane chain, similar to the results of the spectral analysis of the XPS. However, the charge density of the coatings ranged from 3.5×10^{15} to $2.2 \times 10^{16} \text{ N}^+ \text{ cm}^{-2}$, which is all exceed critical threshold of $1.4 \times 10^{14} \text{ N}^+ \text{ cm}^{-2}$. Therefore, all the PDMS-PU-X coatings prepared in this study are expected to exhibit effective anti-bacterial properties.

In general, as the alkane chains of the quaternary ammonium salts increase in length, they tend to curl and fold [49]. Such curling and folding may result in the masking of N⁺ cations, which could lead to a reduction in the charge density on the coating surface [50]. However, the flexibility of the siloxane backbone allows the chains to be easily arranged and rearranged, facilitating a more orderly arrangement of quaternary ammonium salts groups on the surface (Scheme 1 b). This phenomenon is further illustrated by the images in Fig. S6b, which also show that the depth of color on the coating surface decreases with increasing alkyl chain lengths, indicating a reduction in charge density. Overall, PDMS-PU-X coatings prepared in this work are expected to exhibit effective anti-bacterial action. To investigate the effect of charge

density on anti-bacterial properties, surface contact anti-bacterial experiments were conducted.

3.3. Anti-bacterial properties of the PDMS-PU-X coatings

Since PDMS coatings can be adhered to by bacteria under static conditions, it is particularly important to investigate the anti-bacterial properties of PDMS-PU-X coatings. The anti-bacterial activity of the PDMS-PU-X coatings was evaluated used *Gram-negative Escherichia coli* and *Gram-positive Staphylococcus aureus* as the bacterial models.

As shown in Fig. 3a and b, the anti-bacterial efficacy of PDMS-PU-X coatings decreased with increasing alkyl chain length, mirroring the trends observed for charge density and quaternization degrees. While enhanced hydrophobicity (Fig. S7) and longer alkyl chains are generally associated with improved anti-bacterial performance [51], the coatings' efficacy did not scale proportionally with chain length. Notably, PDMS-PU-8 exhibited the highest broad-spectrum anti-bacterial activity, with 98.8 - 99.1 % inhibition against both *E. coli* and *S. aureus*, whereas PDMS-PU-16 displayed the lowest efficacy. These trends align closely with variations in charge density, further supporting its critical role in governing anti-bacterial behavior [52,53]. Furthermore, no inhibition zones were observed around the PDMS-PU-X and PDMS-PU coatings, indicating that the coatings' anti-bacterial properties are contact-based rather than due to the release of anti-bacterial agents (Fig. 3c) [54]. The absence of biocide release helps prevent environmental contamination.

The organisms that contribute to the fouling of PDMS coatings include not only bacteria but also algae, which underscores the importance of investigating the static and dynamic anti-algae properties of PDMS-PU-X coatings.

3.4. Anti-algae properties of the PDMS-PU-X coatings and anti-algae mechanisms

The anti-algae efficacy of PDMS-PU-X coatings was assessed through immersion of coated samples in a suspension containing *Chlorella* and *P. tricornutum* for a period of 14 days. As illustrated in Fig. 4a, the PDMS-PU-X coatings demonstrated a significant anti-algae efficacy in comparison to the unmodified PDMS-PU and uncoated aluminum sheet. Meanwhile, the OD values of algae collected from the surface are shown

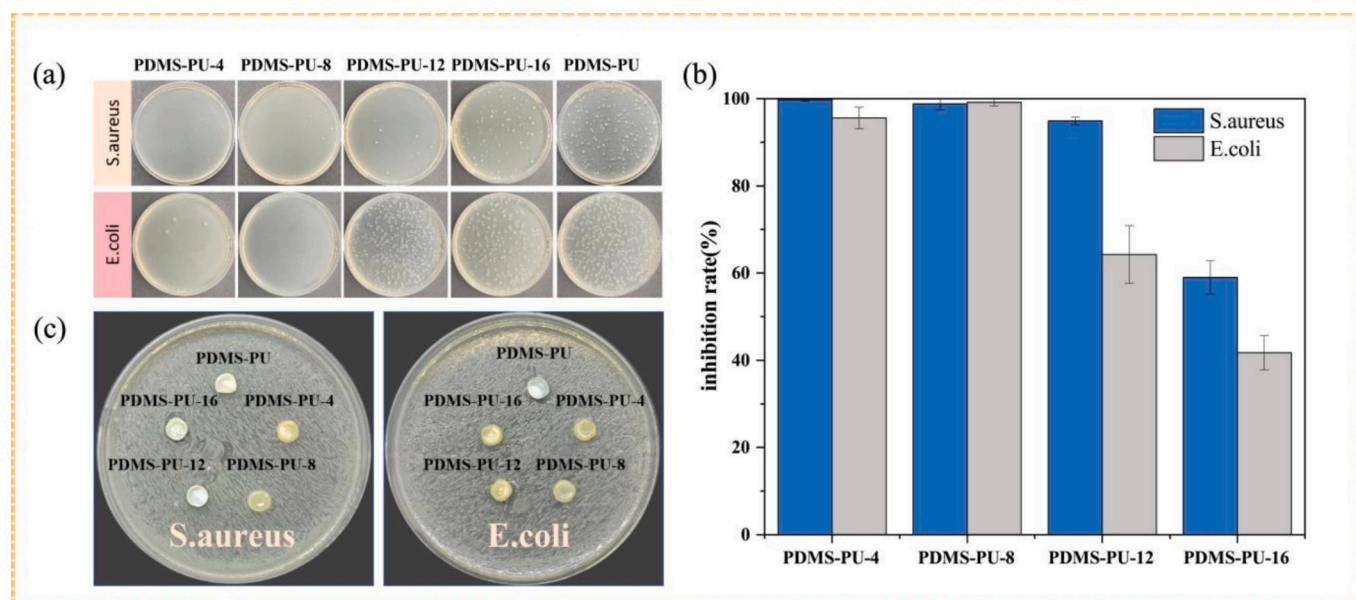


Fig. 3. Anti-bacterial properties of the PDMS-PU-X coatings (a) Images of CFU for *S. aureus* and *E. coli* on agar plates under different conditions (b) Comparison of inhibition rate against *S. aureus* and *E. coli* (c) Zone of inhibition test result of PDMS-PU and PDMS-PU-X.

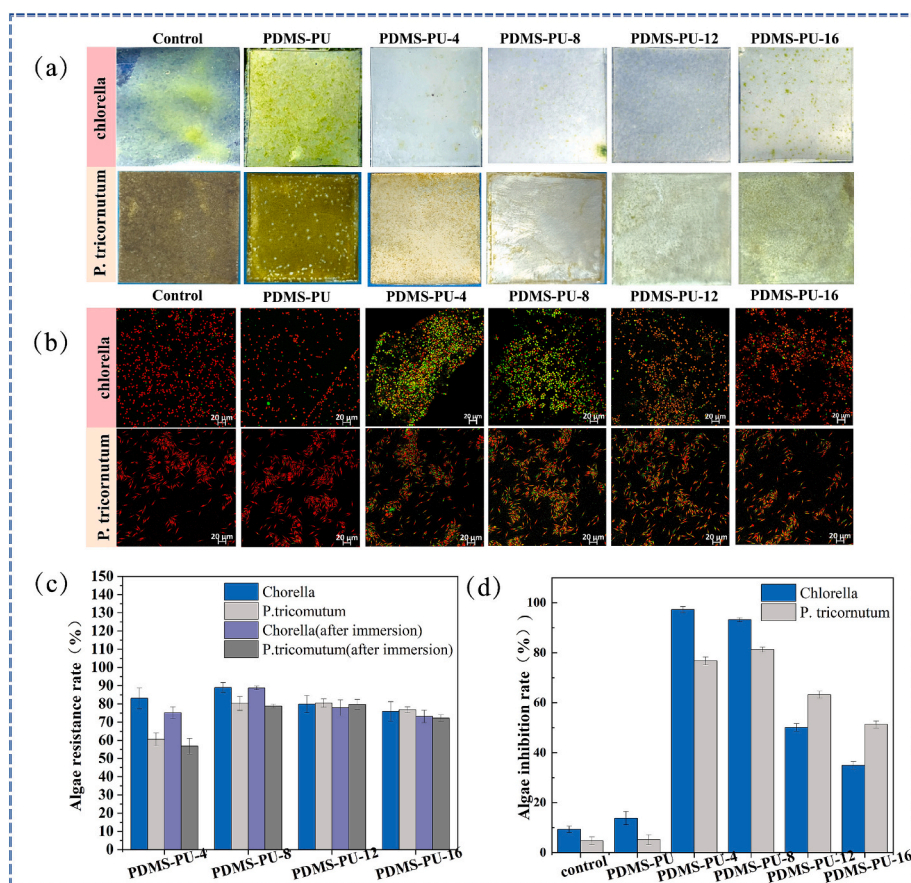


Fig. 4. (a) Images of static anti-algae properties of the PDMS-PU and PDMS-PU-X coatings (the images were captured prior to the rinsing of the coating with DI water), (b) CLSM observation for live/dead staining results of algae adhered on PDMS-PU and PDMS-PU-X coatings, (c) Anti-algae resistance rate of PDMS-PU-X coatings against *Chlorella* and *P. tricornutum*, (d) Algae inhibition rate calculated based on ImageJ software.

in Fig. S8. It can be observed that the amount of algae attached to the PDMS-PU surface is significantly higher than that of the control. This finding is consistent with the previous literature reports indicating that algae tend to preferentially attach to hydrophobic surfaces [55,56]. The anti-algae rate of the coating can then be quantified in terms of the OD value of the algae suspension. The quantitative analysis presented in Fig. 4c indicates that PDMS-PU-8 exhibited the highest anti-algae performance, achieving 88.9 % inhibition for *Chlorella* and 80.2 % for *P. tricornutum*. As illustrated in Fig. 4c, following a 30-day immersion period in a 3.5 wt% NaCl solution, the anti-algae efficacy of the PDMS-PU-X coating exhibited a less pronounced decline, suggesting a durable and prolonged anti-algae capability.

A comparison of the two algae species revealed that the PDMS-PU-X coatings exhibited reduced efficacy against *P. tricornutum* relative to *Chlorella*. This discrepancy is likely due to the siliceous shell component of the *P. tricornutum* cell wall [57]. Penetration of the diatom shell requires longer alkane chains, but also results in a reduction in charge density and algae resistance. Therefore, selection of appropriate alkane chain lengths is required to achieve broad-spectrum anti-algae properties [58]. In this paper, PDMS-PU-8 had the best broad-spectrum anti-algae properties.

Furthermore, CLSM was employed to evaluate the viability of algae that had attached to the coatings. As illustrated in Fig. 4b, the PDMS-PU coating exhibited robust chlorophyll fluorescence (indicating by the red signal) and diminished SYTOX green fluorescence (indicating by the green signal), whereas the PDMS-PU-X coatings demonstrated the inverse pattern. This indicates that the PDMS-PU-X coatings had a more pronounced killing effect on algae, thereby facilitating the removal of dead algae cells from the surface. To provide a quantitative assessment

of the CLSM results, the viable algae ratio (calculated as the area of the green signal divided by the area of the red signal) was determined for the coated samples using the ImageJ software. Fig. 4d illustrates that PDMS-PU-4 and PDMS-PU-8 exhibited the most efficacious killing effects on *Chlorella*, with inhibition rates of 97.3 % and 93.2 %, respectively. This may be attributed to the greater susceptibility of *Chlorella* to cationic charges, with coatings exhibiting high charge density more likely to induce cell death. However, PDMS-PU-4 was not as effective in inhibiting *P. tricornutum*, whereas PDMS-PU-X (X = 8, 12, 16) were more effective in inhibiting *P. tricornutum*. This is likely due to *P. tricornutum*'s higher sensitivity to long-chain quaternary ammonium salts [59]. These findings further indicate the effectiveness of PDMS-PU-X coatings in killing and resisting the adhesion of both *Chlorella* and *P. tricornutum*, thereby enhancing static anti-algae performance.

Since the quaternary ammonium salt primarily acts on the phospholipid bilayer of cell membrane [60], our aim was to gain further insight into the differences in the inhibitory effects of the coatings on *Chlorella* and *P. tricornutum* by monitoring changes in the membrane potential of algae cells using DiBAC4 (3). An increase in fluorescence intensity relative to the control sample indicates cell depolarization, whereas a decrease in fluorescence indicates hyperpolarization [61]. In the course of the experiments, the control group was constituted by algae cultivated on PDMS-PU. As illustrated in Fig. 5, the overall fluorescence intensity of *Chlorella* and *P. tricornutum* on the PDMS-PU-8 coated surface (Fig. 5f and h) was observed to be higher than that of the control (Fig. 5e and g). To quantify the changes in membrane potential, the fluorescence intensity of the CLSM images was analyzed using the ImageJ software. As illustrated in Fig. S9a and Fig. S9b, the mean fluorescence intensity of Fig. 5a is 18.2, while that of Fig. 5c is

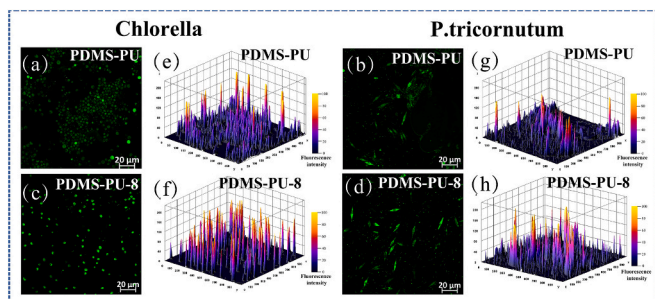


Fig. 5. CLSM observation for DiBAC4 (3) staining results of *Chlorella* (a, c) and *P. tricornutum* (b, d) and their fluorescence thermograms (e, f) *Chlorella* and (g, h) *P. tricornutum*.

42.9, which is 2.3 times that of Fig. 5a. Furthermore, as illustrated in Fig. S9c and Fig. S9d, the mean fluorescence intensity of Fig. 5b is 17.8, while that of Fig. 5d is 26.3, which is 1.48 times that of Fig. 5b.

The findings demonstrated that all algae in contact with the PDMS-PU-8 coatings exhibited cell membrane depolarization. The extent of depolarization was more pronounced in *Chlorella* compared to *P. tricornutum*, indicating that the quaternary ammonium salts on the PDMS-PU-8 coatings had a more pronounced inhibitory effect on

Chlorella. As the depolarization of algae cell membranes alters their permeability, it leads to an accelerated loss of cell contents and eventual cell membrane rupture [62]. To confirm whether depolarization of algae cells occurs in this manner, the surface morphology of algae cells was observed using SEM. Additionally, malondialdehyde (MDA) quantification, DNA leakage detection, and reactive oxygen species (ROS) detection were conducted.

As shown in Fig. 6a, SEM analyses of *Chlorella* and *P. tricornutum* showed that the cellular integrity of the cells exposed to the PDMS-PU-X coating was severely disrupted, whereas the algae cells exposed to the PDMS-PU coating as well as the control group remained intact. This phenomenon is not exclusive to single algae cells, as can be seen in Fig. S10. The observed destruction of cell integrity on PDMS-PU-X coatings may be attributed to the presence of quaternary ammonium salts on the coating surface [63,64]. This irreversible damage to the cell membrane, which ultimately results in cell death, is caused by the strong electrostatic interaction between the positively charged nitrogen atoms of quaternary ammonium salts and the negatively charged cell membrane, as well as the interaction of the long-chain alkyl groups with the phospholipid bilayer of the cell membrane [65].

The exudation of DNA can be employed as an indicator of algae cell rupture on the surface of PDMS-PU-X, thereby confirming the leakage of cellular contents [66]. As illustrated in Fig. 6b, *Chlorella* exhibited a

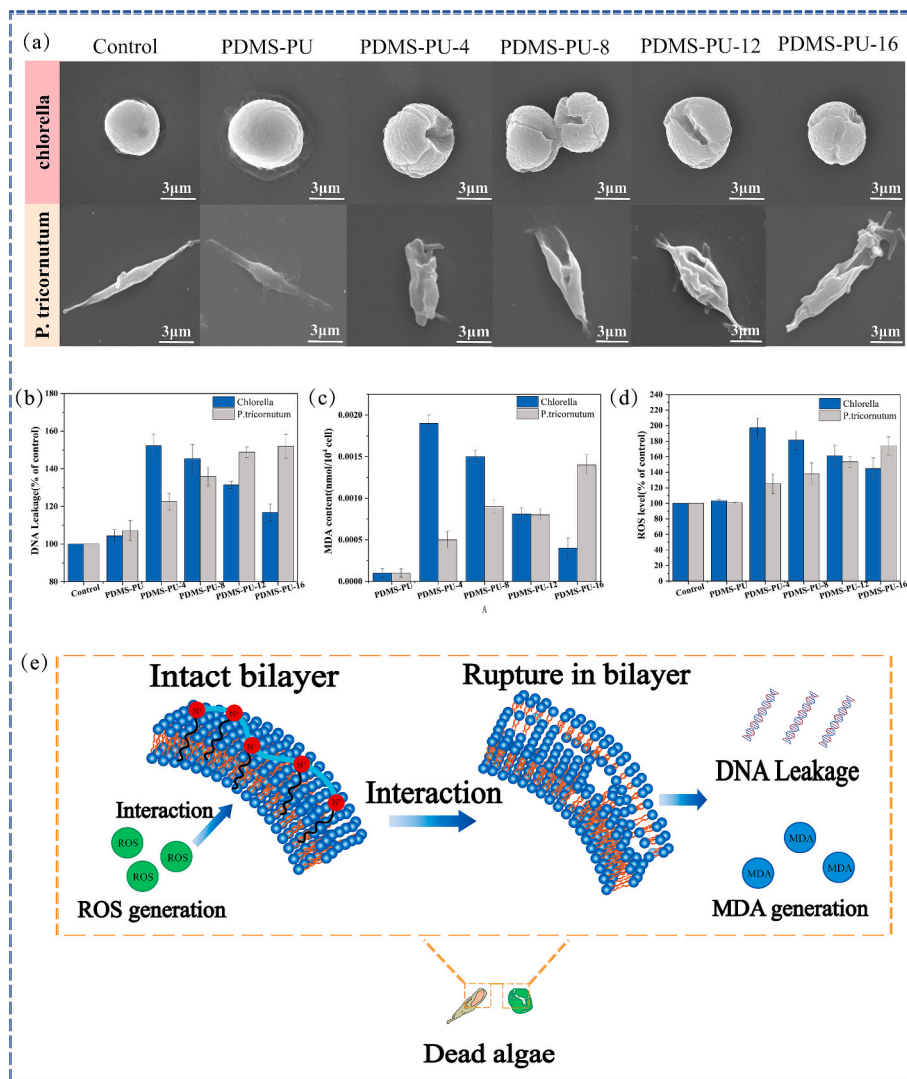


Fig. 6. Effects of PDMS-PU-X coatings on the attached algae. (a) SEM observation of *Chlorella* and *P. tricornutum*. Comparison of (b) DNA leakage from algae cells (c) MDA production, and (d) ROS level in *Chlorella* and *P. tricornutum*. (e) Schematic of the anti-algae mechanism.

declining DNA efflux trend (152.2 % to 114.5 %) with increasing alkyl chain length, whereas *P. tricornutum* showed an opposing trend (127.3 % to 148.1 %). These results suggest an inverse correlation between alkyl chain length and membrane disruption severity in *Chlorella*, contrasting with a direct proportionality observed in *P. tricornutum*. A graph of the specific signal strength data can be seen in Fig. S11.

To gain further insight into the underlying causes of membrane rupture in algae cells, the levels of intracellular ROS and malondialdehyde (MDA) were measured. MDA, a metabolic byproduct of lipid peroxidation, can be triggered by ROS [67,68]. Fig. 6c reveals that MDA content in *Chlorella* decreased with longer alkyl chain length, while *P. tricornutum* displayed a marked increase in MDA accumulation under the same conditions. Consistent with this, ROS levels positively correlated with MDA production (Fig. 6d). The overproduction of ROS induces oxidative stress, which results in cellular damage [69,70]. Collectively, the findings indicate that surface-bound quaternary ammonium salts exhibit stronger inhibition against *Chlorella*, while longer-chain variants demonstrate enhanced efficacy in suppressing *P. tricornutum*. A graph of the specific signal strength data can be seen in Fig. S11.

Based on the aforementioned analyses, the anti-algae efficacy of the PDMS-PU-X coatings arises from interfacial interactions between algal cells and the surface-immobilized cationic compounds. Specifically, the positively charged nitrogen atoms in these compounds electrostatically bind to negatively charged algal membranes, elevating membrane potential and inducing depolarization. This disruption compromises membrane integrity, increasing permeability and triggering DNA leakage. Furthermore, such exposure stimulates excessive intracellular reactive oxygen species production, which initiates lipid peroxidation and malondialdehyde formation. These oxidative byproducts exacerbate membrane damage, ultimately leading to cell lysis and death, as illustrated in Fig. 6e.

These findings provide further insight into the mechanisms by which variations in the alkyl chain length of quaternary ammonium salts affect the anti-algae properties of the coatings. It can be concluded that a balance between alkane chains and charge density is necessary to achieve the optimal broad-spectrum anti-algae properties.

In contrast to commercially available coatings that prevent algae growth by releasing anti-fouling agents, PDMS-PU-X coatings employ a contact-based anti-algae mechanism. This approach ensures prolonged anti-algae performance, offering significant potential for durable marine anti-fouling applications. The anti-fouling properties of the coatings are characterized in the following section, with a view of their practical application in the marine environment.

3.5. Anti-fouling properties of the PDMS-PU-X coatings

The formation of marine fouling is contingent upon the adsorption of algae onto surfaces, rendering resistance to algae adhesion a pivotal indicator of marine anti-fouling performance [71,72]. Consequently, an additional anti-fouling investigation was conducted to determine the anti-algae adhesion properties of the coatings, as illustrated in Fig. 7a. Under static conditions, a considerable number of *Chlorella* and *P. tricornutum* were observed to adhere to the surfaces of the PDMS-PU and PDMS-184 coatings. In contrast, a relatively smaller number of algae were observed to be attached to the surfaces of the PDMS-PU-X coatings. It can also be seen in Fig. S12 that the density of algae attached to PDMS-184 is significantly greater than that of PDMS-PU and PDMS-PU-X, which is due to the fact that algae tend to be more attached to hydrophobic surfaces [56]. Although PDMS-PU-X is also a hydrophobic surface, the quaternary ammonium salts groups on the surface of the coating significantly inhibit the attachment of algae, which can be seen in Fig. 7a [33].

Although PDMS-184 is less resistant to algae adhesion under static conditions, it primarily relies on the shear force of water for fouling removal [73]. Therefore, it is important to examine the ability of the

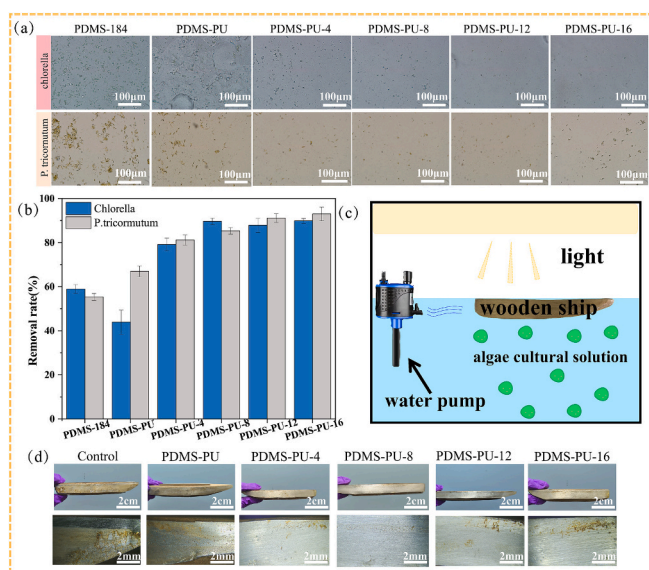


Fig. 7. The anti-fouling performance of PDMS-PU-X coatings. (a) Images of adhered algae on PDMS-184, PDMS-PU and PDMS-PU-X coatings surfaces. (b) The removal rate of coatings (c) Schematic diagram of the simulated ship hull anti-fouling experiment, and (d) Images of attached algae on hull surface.

PDMS-PU-X coatings to remove algae after being flushed with water. As shown in Fig. 7b, the removal rate of the PDMS-PU-X coatings for *Chlorella* and *P. tricornutum* exceeds 80.1 %, reaching up to 91.1 %. The overall algae removal ability of the PDMS-PU-X coatings is higher than that of PDMS-184, indicating that the PDMS-PU-X coatings have good algae removal ability.

To further simulate the practical application scenarios of the coating, an investigation was conducted into the dynamic anti-algae and anti-fouling properties of the coating using a wooden ship model with PDMS-PU-X coatings applied to the hull (Fig. 7c). As shown in Fig. 7d, fewer algae adhered to the surface of the PDMS-PU-X coating, indicating good dynamic anti-algae performance. Meanwhile, it can also be seen from the figure that PDMS-PU-8 has the best anti-algae performance, and no algae can be seen on the surface of the coating, which corresponds to the results of the static anti-algae test (Fig. 4a) and the dynamic anti-algae test (Fig. 7a) of PDMS-PU-8, and indicates that PDMS-PU-8 has the best comprehensive anti-algae performance. The remaining PDMS-PU-4, PDMS-PU-12, PDMS-PU-16 coatings actually have better anti-algae performance than in the static anti-algae experiments, with only a small portion of the coatings having algae attached, suggesting that the coatings have better anti-algae performance when there is water flow. To elucidate the experimental results obtained under simulated conditions, the anti-algae mechanism of the coatings is analyzed as follows.

Since the PDMS coating primarily relies on the shear force of water for fouling removal, it is susceptible to algae attachment and biofilm formation under static conditions. This susceptibility arises from the accumulation of algae and extracellular polymeric substances (EPS), which can lead to further contamination of the coating (as shown in Fig. S13a). Even under dynamic conditions, a certain amount of contamination remained [24]. In comparison, The PDMS-PU-X coatings enhance anti-fouling performance through three key processes (as shown in Fig. S13b). First, the uniformly distributed quaternary ammonium salts groups on the surface of PDMS-PU-X coatings significantly reduced the attachment of algae-like species. Second, the uniformly distributed quaternary ammonium salts groups on the coatings' surfaces kill the attached algae, preventing algae cells from releasing EPS that enhance adhesion. Finally, dead algae cells are more easily washed away by water flow, demonstrating the coating's effective

fouling release performance and dynamic anti-algae properties. The combination of these mechanisms makes the PDMS-PU-X coatings an effective anti-algae solution under both static and dynamic conditions.

To further evaluate the anti-fouling performance of the PDMS-PU-X coatings, a series of anti-fouling experiments were conducted. Fig. 8a illustrates the images of the PDMS-184 and PDMS-PU-X coatings subjected to a variety of fouling liquids, including milk, muddy water, coffee, and a water solution dyed with Food Blue. Iron plates coated with PDMS-PU-X maintain their original cleanliness even after multiple immersions in various liquids.

We conducted the droplets rolling experiments to evaluate the anti-fouling performance of the PDMS-PU-X coatings as shown in Fig. 8b and Video S2. The sliding time of water droplets on PDMS-PU-X is 20 s faster than that of PDMS-184, indicating the low adhesion ability of the PDMS-PU-X coatings to water. Fig. 8c illustrates the self-cleaning processes of the PDMS-PU-X coatings in the presence of solid contaminants. The contaminants could be readily removed by a small quantity of water, indicating that the PDMS-PU-X coatings, with a low surface free energy, exhibited a high water-repellent property. In order to further investigate the superior anti-fouling properties exhibited by the PDMS-PU-X

coatings, we determined the water contact angle of the PDMS-PU-X coatings. As shown in the Fig. S7 and the table S8, the water contact angles of the PDMS-PU-X coatings were $>103.0^\circ$, indicating that the water repellency properties of the PDMS-based silicone materials were retained despite the PU modification and quaternization. In general, the methyl groups of PDMS are positioned at the surfaces and interfaces, which results in low surface free energy. Furthermore, the low surface free energy of PDMS reduces the adhesion strength of macrofouling organisms to the coating. Consequently, these macrofouling organisms can be easily removed from the surface by external shear forces.

The investigation revealed a significant correlation between the real barnacles and the pseudo-barnacles in terms of the bond strength of the coated surface and the dirt release properties of the coated surface [74,75]. The removal strength of the pseudo-barnacle properties of the coatings was examined using an aluminum stud (Fig. 8d). As illustrated in Fig. 8e, the shear forces necessary for the removal of the simulated pseudo-barnacles bonded to the surface of the PDMS-X coating were 0.3–0.4 MPa, which was slightly higher than the 0.3 MPa of the PDMS-184. Nevertheless, the PDMS-PU-X coatings demonstrated a notable capacity for the removal of fouling. The shear force required for removal was

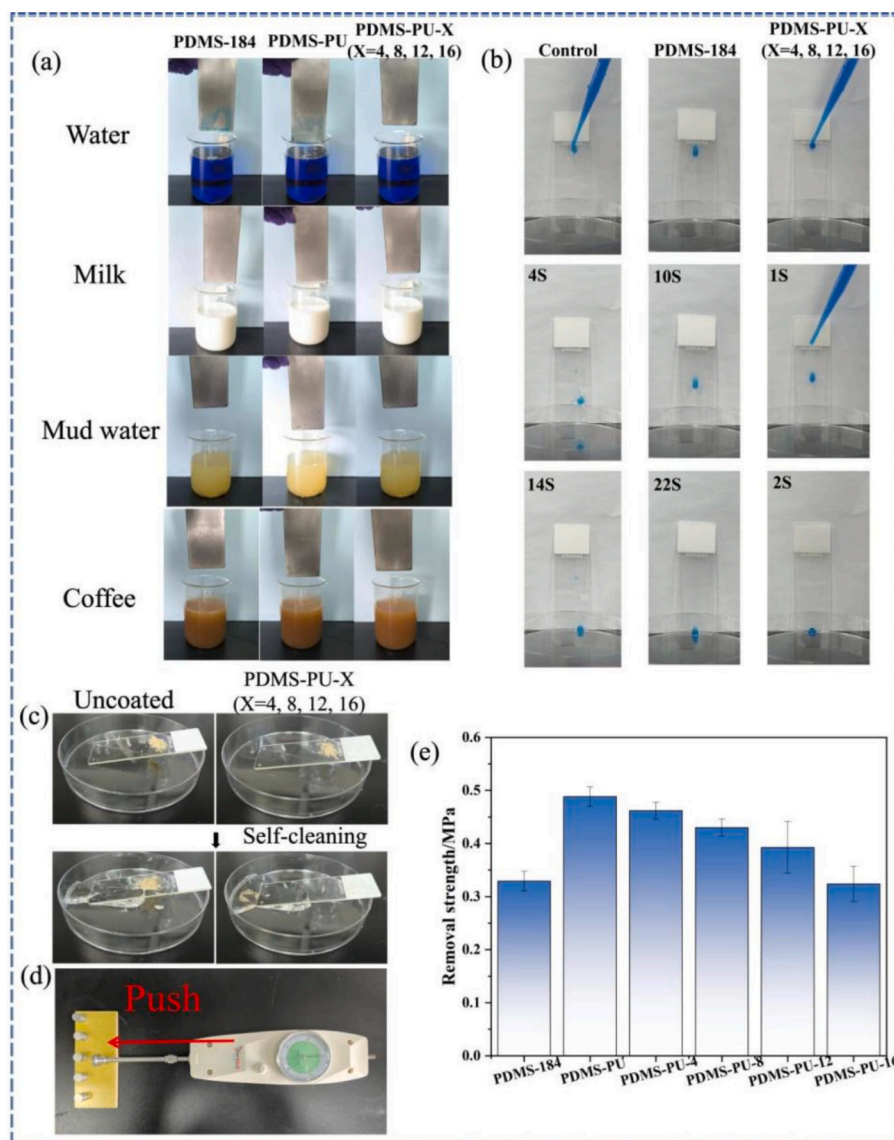


Fig. 8. (a) Anti-fouling performance of PDMS-184 and PDMS-PU-X in four liquid pollutants. (b) Images of droplets rolling down a coated surface. (c) Self-cleaning performance of PDMS-PU-X. (d) Schematic diagram of the pseudo-barnacle shear experiment, and (e) Pseudo-barnacle removal strength of PDMS-184, PDMS-PU and PDMS-PU-X coatings.

observed to decrease notably as the alkyl chain lengths in PDMS-PU-X increased. This phenomenon was found to be closely associated with the static water contact angle on the PDMS-PU-X surface.

Nevertheless, the PDMS-PU-X coatings demonstrated a notable capacity for the removal of fouling. This capacity for the removal of fouling is benefited for increasing the coating's anti-bacterial and anti-algae performance in marine.

3.6. Durability of PDMS-PU-X coatings

PDMS coatings have emerged as a promising solution for environmentally friendly anti-fouling due to their low surface free energy and non-toxic properties. However, these coatings often exhibit from limited durability, mechanical properties, and poor adhesion to substrates. In this work, we developed a polyurethane-modified PDMS to improve the durability and mechanical properties of the coatings.

In order to evaluate the durability of the coatings in practical applications in marine-life situations, the PDMS-184, PDMS-PU and PDMS-PU-X coatings were immersed in 3.5 wt% NaCl solution (as artificial seawater) for 30 days. The surface free energy and water contact angle of the coatings before and after immersing at artificial seawater for 30 days are illustrated in Fig. 9a and b respectively. As illustrated in Table S8, the

surface free energy of PDMS-PU-X coatings exhibits a range of 19.0 (± 1.1) to 21.0 (± 1.5) mJ·m⁻², in comparison to 28.0 (± 1.0) mJ·m⁻² for PDMS-184. According to the literature, organisms exhibit minimal adhesion to surfaces with surface free energy between 20 and 30 mJ·m⁻² [76]. As shown in Fig. 9b, the static WCA of the PDMS-PU-X coatings had a smaller variation, whereas that of the PDMS-PU had a more significant decrease. The improved durability of the PDMS-PU-X coatings in artificial seawater solution may be related with the condensed cross-linking network and hydrophobic alkyl chains (Scheme 1b). The hydrophobic nature of PDMS-PU-X and the robust hydrogen bonding capabilities of the polyurea groups within the polymer network result in an enhancement in the cross-linking density of the polymer network. This effectively impedes the penetration of water molecules into the PDMS-PU-X coatings.

Furthermore, the PDMS-PU and PDMS-PU-X coatings were immersed in diverse pH solutions (pH = 1, 12) for a period of 5 days to evaluate whether the coating surface is still stable. The WCA of the coating in different pH solutions were shown in Fig. S14. Although strong acid, alkali, and NaCl solutions can cause hydrolysis of the PDMS surface [77,78], leading to a decrease in hydrophobicity, the PDMS-PU-X coated surface retains its surface hydrophobicity after sample treatment (Fig. S14c) [79]. The WCA of the coating demonstrated minimal

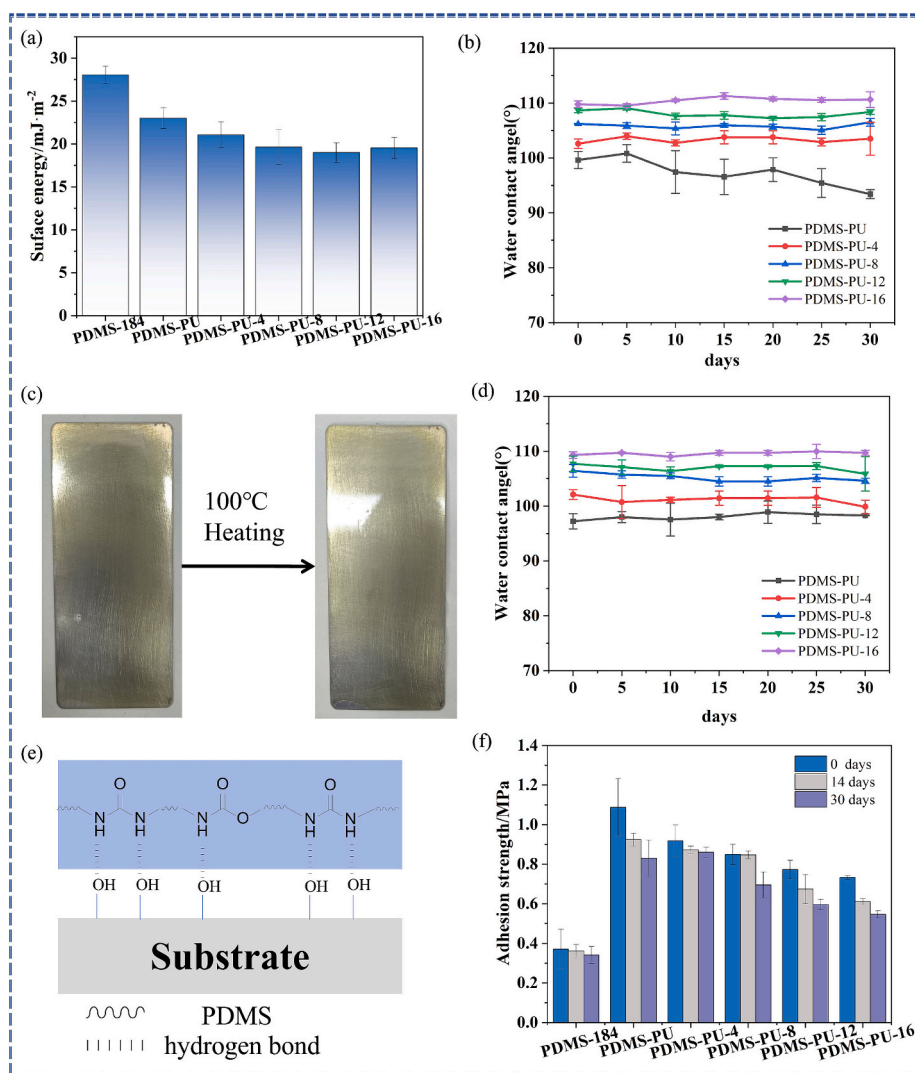


Fig. 9. (a) Surface free energy of PDMS-184, PDMS-PU and PDMS-PU-X coatings surfaces, and (b) Water contact angle of coatings immersion at artificial seawater for 30 days. (c) The images of the sample before and after heating, and (d) Water contact angle of coatings during heating at 100 °C for 30 days. (e) Schematic diagram of hydrogen bonding between PDMS-PU-X coatings and substrate, and (f) Comparing the adhesion strength of the coatings before and after immersion at artificial seawater for 30 days.

alteration, thereby indicating that the coating surface exhibits favorable chemical stability. Finally, the PDMS-PU-X coatings were heated at 100 °C for 30 days to evaluate their thermal stability. The images of the sample before and after heating were shown in Fig. 9c, and water contact angle of the coatings before and after heating were shown in Fig. 9d. No significant changes such as yellowing, discoloration, or peeling of the coating were observed in Fig. 9c after heated at 100 °C for 30 days. Also, the static WCA of the PDMS-PU-X coatings surface the PDMS-PU-X coatings remained almost unaltered over time following heat treatment at 100 °C for 30 days. All of these results indicated that the developed PDMS-PU-X coatings behave effective chemical and thermal stability.

The PDMS-PU-X coatings exhibit durability across different aspects, which is, on one hand, related to the robust Si—O framework structure within the polymer and the coating. On the other hand, the bonding strength between silicone-based anti-fouling coatings and the substrate represents a significant challenge in their practical application. The abundance of urea and carbamate polar groups between PDMS-PU-X allows for strong hydrogen bonding with the substrate, thereby improving adhesion (Fig. 9e). The adhesion of PDMS-PU-X to the substrate reaches 0.7–0.9 MPa (Fig. 9f), which is higher than that of PDMS-184. Following 14 and 30 days immersion period in a 3.5 wt% NaCl solution, a slight decrease in coating adhesion was observed, while the surface morphology remained unchanged (Fig. S15).

The mechanical stability of the PDMS-PU-X coatings was evaluated through a cyclic sand abrasion test (Fig. 10a) [43]. In comparison to the PDMS-184 coating (Fig. 10b), the surfaces of the PDMS-PU-X coatings have not been damaged and continue to demonstrate effective anti-fouling properties and robust substrate adhesion. The presence of a hard segment within the polymer cross-linked network enables the urethane and carbamate groups present in the hard segmented structure to form hydrogen bonds autonomously, thereby maintaining the mechanical properties of the PDMS-PU-X coatings without compromising their dynamic fouling resistance.

These findings further suggest that PDMS-PU-X coatings possess excellent chemical stability and durability due to the introduction of PU and the formation of hydrogen bonding networks, indicating the potential for their use as a marine anti-fouling coating.

3.7. Toxicity assay

Due to the toxicity of conventional commercialized marine biofouling agents, such as biocides, tributyltin, copper-based coatings, the developing eco-friendly and non-toxic anti-fouling agents or

coatings has always been an attractive novel challenge. In order to evaluate the potential toxicity of the PDMS-PU-X coatings to non-target marine organisms, acute toxicity tests were conducted on zebrafish embryos, larvae, and adults. These tests employed commercially available Cu₂O-releasing coatings as a point of comparison. As illustrated in Fig. 11a, zebrafish embryos exposed to the Cu₂O-releasing coating exhibited mortality within 24 h, accompanied by protein coagulation, which impeded normal hatching (Fig. 11d and f). In contrast, zebrafish embryos exposed to the PDMS-PU-X coatings hatched normally into larvae (Fig. 11e and g), indicating that the PDMS-PU-X coatings did not adversely affect the normal hatching process and is essentially non-toxic.

The early-stage fish demonstrate robust responses to contaminants, exhibiting similar sensitivities and a comparable physiological mechanism to that observed in adults. However, the larval stage is the most sensitive, as the embryos possess additional protection through the chorion. Subsequently, the impact of the coating was subjected to further examination utilising zebrafish larvae. As illustrated in Fig. 11h–k, it can be observed that the Cu₂O coating is capable of causing the heart of zebrafish larvae to cease functioning and result in mortality within 24 h (Fig. 11j and Video S3), whereas zebrafish larvae in holes coated with PDMS-PU-X survival (Fig. 11k and Video S4).

Moreover, the acute toxicity of the coatings was evaluated on the adult zebrafish in accordance with the standard (China, GB/T 13267-91). All adult zebrafish exposed to the Cu₂O coating died within three days (Fig. 11l, Video S5), whereas those exposed to the PDMS-PU-X coatings remained alive and exhibited no adverse effects (Fig. 11m, Video S6).

The hatching success rate of zebrafish embryos and the survival rates of zebrafish juveniles and adults were both 100.0 % in the PDMS-PU-X experiment. In contrast, the commercial Cu₂O coating group exhibited a hatching success rate of 0 %, indicating a toxic effect on zebrafish at all developmental stages (Table S9). These findings demonstrate that the PDMS-PU-X coatings are significantly safer and more environmentally friendly than the commercial Cu₂O coating. The PDMS-PU-X coatings developed in this study show great potential for use as an environmentally friendly marine anti-fouling coating. Its excellent durability, non-toxicity to non-target organisms, and effective anti-fouling performance make it a promising candidate for applications in marine environments.

To gain a deeper insight into the performance discrepancies among the PDMS-PU-X coatings and to identify the coating with the optimal overall performance, a comprehensive comparison of their various performance metrics was conducted. These metrics included the anti-algae rate, adhesion strength, pseudo-barnacle removal strength, SFE, and WCA. And the results were listed in Table S8 and Table 1. Evaluating these variables allowed for the selection of the most effective and durable coating for marine anti-fouling applications.

Combining the data from Table 1 and Table 4S, the incorporation of quaternary ammonium salts markedly enhances the static anti-algae properties of the coatings, whereas the introduction of PU chain segment improves their stability, abrasion resistance, and adhesion.

Coating anti-algae performance correlated with charge density and alkyl chain length. PDMS-PU-16 (the lowest charge density) showed the poorest anti-algae effects, followed by PDMS-PU-12, suggesting insufficient charge density compromises anti-algae. Although PDMS-PU-4 had higher charge density, its short alkyl chains failed to penetrate *P. tricornutum*'s thick cell wall (inhibition reduced to 80.2 %). Concurrently, shortened alkyl chains decreased coating hydrophobicity, increasing pseudo-barnacle removal difficulty. PDMS-PU-8 demonstrated optimal balance: 98.8 - 99.1 % antibacterial rate, 88.9 % (*Chlorella*) and 80.2 % (*P. tricornutum*) inhibition rates, with superior anti-fouling and adhesion strength.

Table S10 demonstrates the enhanced performance of PDMS-PU-X coatings compared to literature-reported PDMS modifications. While existing studies rarely assess static anti-algae performance or eco-

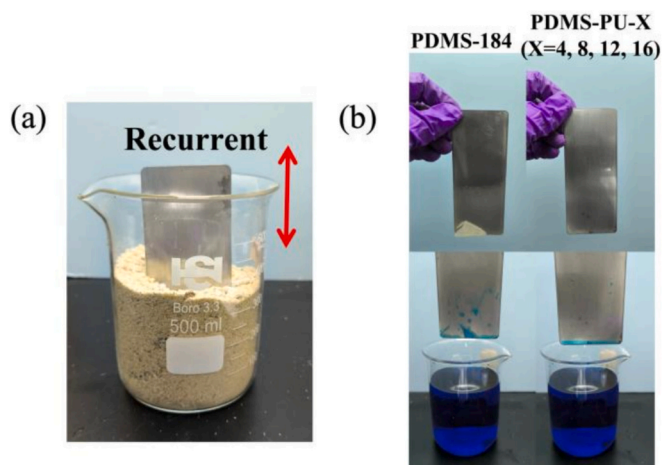


Fig. 10. (a) Schematic diagram of sand cyclic friction, and (b) Comparison of physical images and anti-fouling performance of PDMS-184 and PDMS-PU-X coatings after 100 cycles of sand cyclic friction.

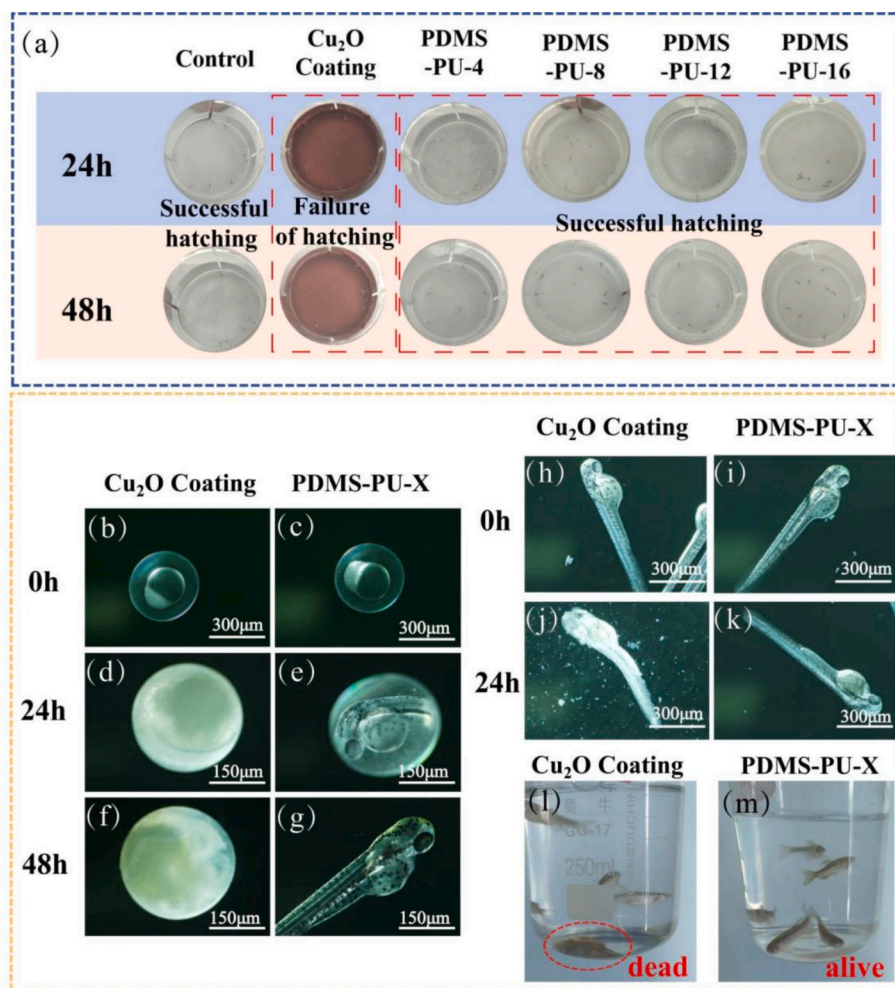


Fig. 11. (a) Images of zebrafish embryo toxicity experiments, (b-g) Microscope images of zebrafish embryo, (h-k) Microscope images of zebrafish juvenile, and (l-m) Images of zebrafish adult.

Table 1

Performance comparison of PDMS-184, PDMS-PU and PDMS-PU-X coatings.

Sample	Anti-algae performance	Anti-fouling performance	Anti-algae adhesion performance	Integrity and stability of coatings	Adhesion strength	Pseudo-barnacle removal strength
Control	–	–	–	/	/	/
PDMS-184	–	+	+	–	+	++
PDMS-PU	–	+	–	+	+++	+
PDMS-PU-4	++	++	+	++	++	+
PDMS-PU-8	+++	++	++	++	++	++
PDMS-PU-12	++	++	++	++	+	++
PDMS-PU-16	+	++	++	++	+	++

Note: " + " means good performance, more " + " indicates better performance, "–" means bad performance.

toxicity, our PU-modified PDMS incorporated with quaternary ammonium salts achieved a 99.7 % antibacterial rate. Notably, PDMS-PU-8 exhibited unprecedented static anti-algae inhibition (~90.0 %, unreported elsewhere) and retained good anti-algae properties after 30-day NaCl immersion (Fig. 4c), attributed to hydrogen bonding from urea and carbamate groups that enhance substrate adhesion. The synergistic design combines: (1) PU to wear resistance and durability, (2) quaternary ammonium salts for dynamic/static anti-algae and bacterial properties, and (3) PDMS as the main chain can reduce the removal strength of pseudo-barnacles. This tripartite integration delivers durable, eco-friendly anti-fouling functionality.

4. Conclusion

In the study, a novel environmentally friendly polyurethane-modified PDMS coatings with dual anti-bacterial, anti-algae, and anti-fouling properties were prepared. The hard segment (formed with MDEA and IPDI), flexible parts (long PDMS chains and alkane chains in quaternary ammonium salts) combined with strong cross-linked hydrogen bonds ensure the PDMS-PU polymer coatings in hydrophobic, and enhanced the coatings properties of mechanical, durability, and static anti-algae properties. The formation of a covalent bond between a quaternary ammonium salts (QAS) and PDMS significantly improves the

coating's ability to resist algae growth. This research examines the relationship between the coating's anti-algae efficacy and the structural characteristics of the quaternary ammonium salts, as well as the underlying mechanisms. The PDMS-PU-8 demonstrated the most favorable overall performance. The anti-algae mechanism is primarily due to the interaction between the quaternary ammonium salts's alkane chains and the cell membranes of algae, which results in depolarization of the latter, increased levels of ROS, and irreversible damage to the cell membranes. This ultimately results in the leakage of cell content and cell death. Moreover, the study identifies a notable correlation between the coating's charge density and the alkyl chain lengths in the quaternary ammonium salts, with longer alkyl chain lengths reducing charge density and anti-algae performance. The alkyl chain lengths of eight carbon atoms has been identified as the optimal length for achieving the desired anti-algae efficacy. Furthermore, the PDMS-PU-X coatings have been demonstrated to be non-toxic to non-target organisms, such as zebrafish, and exhibit environmental compatibility. This study not only provides a new idea for the development of environmentally friendly and efficient marine anti-fouling coatings, but also provides a theoretical basis for the structural design and performance optimization of quaternary ammonium salt-modified coatings.

The following files are available free of charge. Algae culture solution components; Scheme of Synthesis of PDMS-PU-X; GPC data of polymers; ^1H NMR and ^{13}C NMR Characterization of the PDMS-PU-X; UV/vis spectra of PDMS-PU-X; Specific data on the charge density and DQ of the coating; WCA of PDMS-PU-X; Specific data of coatings; Fluorescence intensity of algae; SEM images of coatings; Survival rate of zebrafish. The specific rinsing procedure (Video S1). Droplets rolling down on the PDMS-184 and PDMS-PU-X coatings surface (Video S2). The heart of zebrafish larvae to cease functioning and result in mortality within 24 h (Video S3). The heart of zebrafish larvae still beating within 24 h (Video S4). All adult zebrafish exposed to the Cu_2O coating died within three days (Video S5). All adult zebrafish exposed to the PDMS-PU-X coatings were alive and exhibited no adverse effects (Video S6). Supplementary data to this article can be found online at <https://doi.org/10.1016/j.surfcoat.2025.132211>.

CRediT authorship contribution statement

Zhuorui Wu: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Xiang Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Zhifeng Hao:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Xiaonan Huang:** Investigation, Data curation. **Jiada Cui:** Investigation, Data curation. **Hongyu Yang:** Investigation, Data curation. **Mohamed S. Selim:** Resources, Methodology, Investigation, Funding acquisition. **Jian Yu:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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